

Predicting missing data based on multi-site wearable sensor data

Master's Programme in Computer, Communication and Information Sciences

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Abstract

Human motion tracking and human activity recognition technology have significant applications in medical rehabilitation, virtual reality, and sports science. However, its deployment faces two main challenges: 1) sensor data collection can be disrupted by environmental interference, causing data loss or quality fluctuations; 2) traditional motion capture systems, which rely on dense sensor networks, negatively impact wearability and energy efficiency. To address these issues, this study explores several deep-learning-based frameworks for sensor data generation. The frameworks

aim to accurately predict missing motion data from other sites' sensor information, supporting lightweight motion capture systems. The study reviews sensor data imputation and generation methods, comparing four deep-learning models: 1) convolutional autoencoder (AE) for feature reconstruction; 2) variational autoencoder (VAE) with probabilistic modelling; 3) convolutional neural network combined with long short-term memory (CNN-LSTM) for spatiotemporal feature fusion; 4) conditional generative adversarial network (cGAN) with adversarial training. To assess the impact of sensor topology on prediction, two experimental scenarios are designed. The first scenario uses a global mode with all non-target site sensor data. The second one uses a local mode with only neighbouring site sensor data of the target site. The experiments use a multi-metric evaluation system. Results show that in global sensing mode, CNN-LSTM, with hierarchical spatiotemporal feature extraction, performs best in loss control, temporal correlation, and signal fidelity. cGAN's accuracy drops due to adversarial training induced uncertainty. In local sensing mode, VAE shows more robustness, but all models' absolute performance declines significantly, indicating limited predictive capability with neighbouring node data. The study finds: 1) Discriminative models (e.g., CNN-LSTM) excel in deterministic prediction with sufficient global data by explicitly modelling spatiotemporal dependencies; 2) Generative models (e.g., VAE) are more adaptable to local data (smaller datasets) due to their probabilistic representation; 3) Model performance is highly sensitive to sensor topology, requiring dynamic architecture selection based on sensor density. These findings offer theoretical support for missing-sensor prediction and lightweight motion capture systems. Future research should further explore the impact of sensor node spatial topology on generation.

Keywords Human Motion Tracking Technology, Human Activity Recognition, Missing Sensor Data Generation, AE, VAE, CNN-LSTM, cGAN

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Abbreviations

AE

CNN-LSTM

Autoencoder

Convolutional Neural Network combined with Long Short-Term Memory

cGAN

DAE

GANs

GRUs

HAR

HMT

IMUs

KNN

LSTM

Conditional Generative Adversarial Networks

Denoising Autoencoder

Generative Adversarial Networks

Gated Recurrent Units

Human Activity Recognition

Human Motion Tracking

Inertial Measurement Units

K-Nearest Neighbors

Long Short-Term Memory

MAE

MSE

missForest

Mean Absolute Error

Mean Squared Error

Random Forest-based Imputation

NNRW Neural Networks with Random Weights

missForest

NNRW

RMSE

SNR

VAE

Random Forest-based Imputation

Neural Networks with Random Weights

Root Mean Squared Error

Signal-to-Noise Ratio

Variational Autoencoder

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1 Introduction

1.1 Background & Motivation

Human motion tracking technology has significant applications in medical rehabilitation, virtual reality, and sports science. In rehabilitation, it allows for precise monitoring of patients' progress, helping doctors and therapists tailor treatment plans [1]. In virtual reality, it enables immersive interaction by synchronizing users' real-world motions with their virtual avatars [1]. In sports science, it aids in optimizing athletes' biomechanics through motion analysis [1]. Similarly, Human Activity Recognition (HAR) plays a crucial role in smart healthcare, human-computer interaction, and ambient assisted living by identifying human behaviours from motion data [7].

However, both applications face a common challenge: the reliability of motion data is often compromised by sensor data loss or quality degradation, which can be caused by environmental interference, limited sensor coverage, or energy constraints. Traditional motion capture systems, which rely on dense sensor networks, not only increase wearability burden and energy consumption but also make the system vulnerable to partial sensor failure. Existing commercial motion tracking systems, like XSens MVN, face challenges. They require up to 17 inertial measurement units (IMUs) for high-precision motion reconstruction, but this dense sensor

deployment has issues [1]. Sensor signals are easily affected by electromagnetic noise, occlusion, and hardware faults, leading to data loss or quality fluctuations. Indoors, electronic devices can interfere with sensors; outdoors, objects like buildings can block signals. These problems reduce system reliability and affect application accuracy.

Moreover, dense sensor networks decrease wearability and increase energy consumption. In rehabilitation, bulky sensors can restrict patients' movements and affect recovery [1]. In virtual reality, frequent data transmission causes delays, disrupting the user experience [1].

Recent deep learning advancements offer solutions by generating missing data through multi-part correlations. For example, deep learning models can capture the coordination patterns of different body parts during motion. Yet, current research has limitations. Sensor topology sensitivity is often overlooked. The spatial layout of sensors impacts prediction accuracy, but there's a lack of systematic analysis on this relationship. Also, there's no clear framework for comparing discriminative models (e.g., CNN-LSTM) and generative models (e.g., VAE, cGAN) across different data scenarios. This makes model selection challenging in practice.

Addressing these issues is vital for advancing motion tracking technology and human activity recognition. Research on sensor layouts can optimize deployment, improving data reconstruction accuracy while reducing sensor

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numbers. Systematic model comparisons can guide selection, enhancing motion capture system performance. This will support the development of lightweight, robust motion tracking systems.

1.2 Objectives

This study aims to address the missing data challenges of motion sensors in practical applications through four key research initiatives:

1.2.1 Deep Learning-Based Sensor Data Generation Framework

Human motion tracking technology holds significant value in fields such as medical rehabilitation, virtual reality, and sports science. However, its practical deployment faces numerous challenges. Sensor data collection is susceptible to environmental interference, leading to data loss or fluctuating quality. Additionally, traditional motion capture systems rely on dense sensor networks, which negatively impact wearability and energy efficiency. To address these issues, this paper proposes a deep learning-based sensor data generation framework. The framework aims to accurately predict missing motion data from information at other sensor sites, thereby supporting the development of lightweight motion capture systems. By leveraging deep learning models to capture the spatiotemporal correlations in human motion, the framework enhances the robustness and reliability of motion tracking systems under conditions of sensor deficiency.

1.2.2 Comparative Analysis of Four Model Architectures

Under two perception paradigms—global perception (utilizing data from allbody sensors) and local perception (relying only on information from adjacent nodes)—this paper systematically compares the performance of four representative deep learning models: Convolutional Autoencoder (AE), Variational Autoencoder (VAE), Convolutional Neural Network combined with Long Short-Term Memory (CNN-LSTM), and Conditional Generative Adversarial Networks (cGAN). Through comparative studies, this paper reveals the strengths and limitations of each model in different data availability scenarios, providing important empirical basis for model selection in practical applications. The results show that CNN-LSTM performs best under the global perception mode, while VAE demonstrates stronger robustness under the local perception mode.

1.2.3 Sensor Deployment Optimization Analysis

Through experimental research and theoretical analysis, this paper investigates the impact of sensor density and spatial topology on model performance in the context of missing sensor data prediction. The study identifies that model performance is highly sensitive to the spatial arrangement and availability of sensor data. Specifically, under the global sensing mode, where all non-target sensors across the body are used to predict the missing data from a target sensor, increasing sensor density is shown to significantly

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improve prediction accuracy. In contrast, under the local sensing mode, which relies only on data from sensors spatially adjacent to the target sensor, the predictive capability of all models declines substantially. This highlights the critical limitation of local information in reconstructing missing motion signals and underscores the importance of global context in achieving robust predictions. These findings provide valuable theoretical insights for optimizing sensor deployment strategies, enabling a better trade-off between system complexity (e.g., number of sensors, energy consumption) and tracking accuracy.

By advancing the fundamental understanding of sensor data interaction in deep learning models, this paper not only offers a new perspective for theoretical research but also provides practical implementation guidelines for real-world applications. These research findings will help promote the wide-spread application of motion tracking technology and human activity recognition in fields such as medical rehabilitation, virtual reality, and sports science.

1.3 Delimitations

This study explicitly defines its scope and boundaries to ensure methodological rigor and focused outcomes:

1.3.1 Focus on Periodic Motions

The experimental scope prioritizes periodic movements (e.g., walking, running) while excluding non-periodic or sudden actions. Periodic motions exhibit relatively stable kinematic patterns and regularities, facilitating effective learning and prediction by deep learning models. In contrast, non-

peri-odic or abrupt movements involve complex and variable dynamics, demand-ing more sophisticated modelling approaches, which are reserved for futureresearch phases.

1.3.2 Sensor Type Restriction

The investigation is limited to accelerometer data, excluding multimodal sen-sor inputs such as gyroscopes. While accelerometers provide critical linearacceleration measurements essential for motion capture, multimodal sensors(e.g., gyroscopes for angular velocity) could enhance motion characterizationat the cost of increased data processing complexity and computational over-head. This methodological choice prioritizes foundational insights from sin-gle-modality acceleration data, with multimodal integration planned as asubsequent research extension.

1.3.3 Offline Training Emphasis

Model optimization primarily employs offline training methodologies to en-sure robust parameter tuning using comprehensive datasets. Although of-fline training guarantees model stability and performance refinement, real-

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time performance validation—critical for practical deployment—remains asubject for future investigation. Subsequent phases will address real-time in-ference capabilities after establishing baseline model efficacy.

These scope definitions aim to balance research depth with feasibility, ena-bling systematic advancements while delineating clear pathways for futureexpansions.

1.4 Outlines

The research is organized into five chapters:

Chapter 2 Literature Review: This chapter reviews existing sensor data im-putation methods (e.g., statistical interpolation, traditional ML approaches)and deep learning applications in motion synthesis, focusing on spatiotem-poral data modelling, establishing theoretical foundations for selecting AE,VAE, CNN-LSTM, and cGAN as experimental models.

Chapter 3 Methods and Implementation: This chapter proposes four deeplearning models (AE, VAE, CNN-LSTM, cGAN) adapted for motion data im-putation, defines two sensor topology modes: global mode (all non-targetsensors as input) and local mode (only neighbouring sensors of the target sitesas input) and introduces evaluation metrics to assess prediction accuracy andsignal quality.

Chapter 4 Results: This chapter compares model performance across globaland local perception paradigms, analysing how different architectures han-dle varying data availability. Experiments evaluate four models (AE, VAE,CNN-LSTM, cGAN) using RMSE, MSE, MAE, R^2 , CC, and SNR metrics. Re-sults reveal CNN-LSTM excels in global mode with superior error control andSNR, while VAE demonstrates robustness in local mode despite performancedrops. cGAN shows

instability under low-data conditions, and all models de-grade significantly with reduced sensor inputs, highlighting the need for adaptive architecture selection based on data density.

Chapter 5 Summary: This chapter discusses design implications for light-weight systems, emphasizing CNN-LSTM's suitability for global mode and VAE's adaptability in local-mode environments. It identifies limitations in GAN's adversarial training stability and highlights probabilistic modelling's role in data-scarce contexts. Future work focuses on optimizing model architectures, improving real-time performance, expanding to non-periodic motions, and refining sensor deployment strategies to advance wearable motion capture technology.

2 Literature review

The continuous advancement of wearable technologies has significantly expanded their applications across healthcare, motion analysis, and behavioural tracking. Sensor-collected data enable researchers to perform real-time analyses of human motion, health status, and environmental interactions. However, these devices frequently encounter sensor data loss caused by hardware malfunctions, connectivity disruptions, and energy constraints. Such missing data critically compromise the accuracy and reliability of downstream tasks, including activity recognition, health monitoring, and personalized recommendation systems. Given this context, researchers have increasingly focused on developing advanced data imputation and prediction techniques to address sensor data incompleteness.

This section reviews three research domains central to the current study:

- (1) Types and applications of wearable sensors;
- (2) Definitions of human activity recognition, human motion tracking and data characteristics of motion sensors;
- (3) Methods for predicting missing sensor data, encompassing traditional interpolation approaches, data fusion strategies, and deep learning-based predictive models.

2.1 Wearable Sensors

Wearable sensors represent a class of electronic devices that integrate advanced sensing technologies with wearable design principles. Through direct contact with the human body or integration into wearable systems, these sensors enable accurate and real-time detection of diverse data types, including physiological parameters, motion states, and environmental information. The collected data are subsequently converted into electrical signals or other processable formats for transmission and analysis [3].

In contemporary society, wearable sensors demonstrate extensive application prospects and significant value across multiple domains. Within healthcare, they play a pivotal role in disease monitoring and diagnosis by enabling continuous tracking of vital signs (e.g., heart rate, blood oxygen levels) and biomarkers. Additionally, during rehabilitation processes, they provide critical

support for monitoring motor function recovery and delivering real-time feedback to patients [4]. In sports science, these sensors facilitate real-time motion state monitoring and training optimization. By analysing technical movements, they identify potential flaws in motion patterns or injury risk factors, thereby enhancing athletic safety and performance [5]. In daily life, wearable sensors contribute to health management by tracking activity levels, sleep quality, and other behavioural metrics, empowering users to gain insights into their physical activity patterns and adjust lifestyles

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accordingly [6]. Furthermore, advancements in smart technologies have endowed wearable sensors with intelligent interaction capabilities, enabling seamless connectivity with other smart devices or platforms to deliver personalized health management solutions and enhanced user experiences [6].

Wearable sensors can be categorized into three primary types based on their measurement objectives, each focusing on distinct data modalities to provide comprehensive health, motion, and environmental insights:

2.1.1 Physiological Parameter Sensors

Specialized in monitoring users' physiological states and health-related metrics, such as heart rate sensors, blood oxygen sensors, and body temperature sensors.

2.1.2 Motion Parameter Sensors

Designed to capture and analyse motion-related data for understanding movement patterns and biomechanical efficiency. Common examples include accelerometers, gyroscopes, and magnetometers.

2.1.3 Environmental Sensors

Focused on detecting ambient conditions to enhance quality of life and safety. Typical variants include light sensors, sound sensors, and barometric pressure sensors.

Compared to other sensor categories, motion sensors exhibit particularly broad applicability. As outlined above, motion sensors—a subset of wearable devices—are specialized electronic components capable of sensing and recording human movements. By capturing dynamic body motion information, they generate time-series data critical for applications ranging from activity recognition to biomechanical analysis.

Table 1 Presentative Motion Parameter Sensors Summary

Typical motion sensors	Sensor Data	Function	Application
------------------------	-------------	----------	-------------

Accelerometer sensor			
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Acceleration values are usually provided along the three axes of X, Y, and Z. The value of each axis represents the acceleration along that direction, with the unit being meters per second squared (m/s^2).

By analyzing these acceleration values, motion states such as walking, running, and standing can be identified. For example, the acceleration graph will show periodic fluctuations when walking.

Accelerometer data can be used to detect changes in a user's posture, activity levels, and even specific events such as falls.

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Gyroscope sensor

Angular velocity, which reflects the rate at which an object rotates around each axis, is also usually provided along the three axes of X, Y, and Z, with the unit being degrees per second ($^\circ/\text{s}$).

When it is used in combination with an accelerometer, it can help calculate precise postures and angle changes.

It is used for posture detection, such as the tilt of the torso, the rotation of the arms or the turning of the head, etc.

Magnetometer sensor

The magnetic field strength is usually given along the three axes of X, Y, and Z, and its unit is microtesla (μT).

It provides directional information by measuring the strength of the geomagnetic field and can be used to detect the orientation of the device.

Navigation and direction detection. It is often used in combination with accelerometers and gyroscopes to provide a complete sense of direction.

Gravity sensor

The direction and intensity of gravity. It detects the relative gravity direction of the device and is provided along the X, Y, and Z axes.

By analyzing the gravity data, it's possible to determine whether the device is placed horizontally or vertically, or the tilt angle in a certain direction.

It is used for posture recognition, especially for the posture of the device when it is stationary.

Linear acceleration sensor

Linear acceleration, excluding the influence of gravitational acceleration.

It can better reflect the actual motion state without being affected by the device's orientation.

It is mainly used to identify more precise motion patterns, such as instant acceleration or deceleration during running.

2.2 Why Sensor Data Generating is Important?

Human motion tracking (HMT) and Human Activity Recognition (HAR) are two closely related areas of human motion perception technology, both aiming to understand and interpret human movements using various sensing modalities. HMT refers to the process of estimating and reconstructing the position, orientation, and movement of human body parts in real time, often

with the goal of capturing detailed kinematic information such as joint angles, limb trajectories, and motion dynamics [61]. It has broad applications in fields such as medical rehabilitation, virtual reality, sports science, and animation, where precise motion data is essential for analysis, interaction, or simulation [61]-[63]. Vision-based HMT methods leverage cameras and computer vision algorithms (e.g., OpenPose [64], MediaPipe [65]) to track human poses in 2D or 3D space,

while sensor-based approaches rely on wear-able inertial measurement units (IMUs) to directly measure motion dynamics with high temporal resolution. Hybrid systems that combine vision and inertial sensors have also been shown to improve robustness under occlusion or lighting variations [66].

Similarly, Human Activity Recognition (HAR) is the process of identifying and classifying human activities through various technical means, with the primary goal of determining specific actions—such as walking, running, or sitting—using sensor data [7]. It has emerging applications in diverse fields, including health monitoring systems, surveillance and security, gaming, and virtual reality [8]. HAR systems are broadly categorized into wearable, non-wearable, and hybrid systems [9][10]. Wearable HAR systems integrate motion sensors (e.g., accelerometers, gyroscopes, magnetometers) into embedded devices worn on specific body positions (e.g., wrists, waist, neck) to recognize daily activities [11]. Non-wearable HAR systems employ ambient sensors (e.g., cameras, pressure sensors, acoustic sensors, RFID) deployed in the environment to monitor human actions [10][12]. Hybrid HAR systems combine wearable and non-wearable sensors to achieve higher detection accuracy [15]. Among these, wearable HAR systems are the most widely adopted due to their portability, cost-effectiveness, and superior performance [14].

Both HMT and HAR rely heavily on the quality and quantity of sensor data for accurate and reliable performance. Missing or noisy data can significantly degrade the effectiveness of motion tracking algorithms and lead to misclassification in activity recognition tasks [67][68]. This shared dependency on sensor data motivates the development of robust data imputation and generation techniques to enhance the performance of both HMT and HAR systems under real-world conditions.

2.2.1 Suffering from missing values

- (1) Electromagnetic or environmental interference disrupting sensor operations.
- (2) Hardware malfunctions during data acquisition or transmission.
- (3) User discomfort leading to intermittent sensor removal, particularly during sleep or work.
- (4) Technical constraints, such as limited battery life, synchronization issues, or software crashes [16]-[26].

Missing data significantly degrade HAR or HMT performance, as traditional approaches (e.g., deletion or last-value imputation) reduce dataset integrity and model generalizability [25]. To address this, synthetic data generation has emerged as a robust strategy. By leveraging existing data patterns and deep learning algorithms, synthetic data can fill gaps while preserving statistical distributions, thereby enhancing data continuity and downstream task reliability.

2.2.2 Advantages of synthetic data generation

- (1) Enhanced System Robustness

In real-world scenarios, data loss is pervasive due to hardware limitations (e.g., weak Wi-Fi signals, packet collisions) or user behavior (e.g., sensor displacement) [17][26]. Conventional methods like data deletion or naive imputation impair machine learning models by reducing training samples or introducing bias. Synthetic data reconstruction maintains temporal continuity, enabling stable operation of critical systems (e.g., health monitors or industrial sensors) even during data gaps. For instance, in motion analysis, reconstructing missing limb sensor data preserves gait pattern integrity, while in healthcare, imputed physiological parameters ensure accurate diagnostic insights.

(2) Reduced Hardware Dependency and Energy Optimization

Predictive models for sensor data synthesis reduce reliance on dense physical sensor deployments, lowering hardware costs and maintenance efforts [27][28]. In IoT networks, selectively deactivating sensors during low-priority periods—while relying on predictive models—can optimize energy consumption without sacrificing data fidelity. This approach is particularly valuable in resource-constrained environments, such as wearable health monitors requiring prolonged battery life.

(3) Anomaly Detection Support

Synthetic data generation establishes baseline expectations for sensor outputs. Significant deviations between predicted and actual values may indicate sensor faults, system failures, or environmental anomalies. This capability enhances predictive maintenance and fault diagnosis, enabling early detection of critical issues.

By addressing data incompleteness, synthetic data generation not only improves system reliability but also optimizes resource utilization and expands analytical capabilities across applications.

2.3 Missing Data Prediction Methods

Missing sensor data can be categorized into three primary patterns:

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(1) Missing at Random - Data loss occurs in arbitrary temporal sequences without systematic regularity.

(2) Periodic Missing - Data gaps follow non-random, cyclical patterns (e.g., intermittent sensor deactivation).

(3) Complete Loss - Data is entirely absent due to critical failures such as battery depletion or system malfunctions [21][27].

The third pattern, often caused by hardware limitations or abrupt technical issues, is characterized by complete data unavailability over specific intervals. Addressing these missing data scenarios requires tailored prediction strategies, ranging from conventional statistical methods to advanced machine learning algorithms. This section systematically reviews these approaches, progressing from traditional techniques to contemporary data-driven solutions.

2.3.1 Traditional methods

Traditional methods rely on time-series features and statistical principles, primarily suitable for small-scale missing data scenarios. Their technical approaches can be categorized as follows:

(1) Interpolation Methods

These methods estimate missing values by constructing functional relationships between known data points. Key techniques include:

Linear Interpolation: Assumes linear variation between adjacent data points, offering computational simplicity but failing to capture nonlinear trends.

Spline Interpolation: Fits curves using piecewise polynomials, achieving high accuracy for short continuous missing segments.

Polynomial Interpolation: Approximates data distribution via high-order polynomials, but risks overfitting oscillations at missing interval edges.

Limitations: Reliant on the assumption of local data continuity and sensitive to long-term missing data or high-noise conditions.

(2) Statistical Estimation Methods

Table 2 Description of Statistical Estimation Methods

Methods	Description
---------	-------------

Mean Imputation	
-----------------	--

a. Direct Mean Imputation:	Replaces missing values with the global mean, leading to underestimated variance and distributional bias.
----------------------------	---

b. Moving Average Method:	Computes local means (e.g., exponentially weighted moving average) within sliding
---------------------------	---

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windows to mitigate global mean bias. However, it exhibits lag effects for long-term missing data [24] [25].

Similarity-Based Imputation

- a. Hot-Deck Method: Matches records with similar features in the current dataset to fill gaps, but suffers from the "curse of dimensionality" in high-dimensional data.
- b. Cold-Deck Method: Relies on strict rule-based matching from predefined external reference datasets, requiring additional maintenance costs and lacking generalizability.

Probabilistic Iterative Methods

Maximum Likelihood Estimation: Optimizes parameters of latent variable models iteratively via the Expectation-Maximization algorithm. Computational complexity escalates rapidly with data scale and missing ratio, often causing convergence failures in large-scale missing scenarios [32].

Other Statistical Techniques

Include regression imputation (predictive modeling using covariates), Bayesian estimation (robustness enhancement through prior distributions), and deletion methods (e.g., listwise or pairwise deletion). These approaches require predefined data distribution assumptions and struggle with high-dimensional sparse data [29][30].

Traditional methods excel in handling small-scale continuous missing data but face critical limitations: (a) Dimensional Sensitivity: Similarity metrics and statistical assumptions break down in high-dimensional spaces; (b) Long-Term Gap Handling: Interpolation and mean-based methods fail to model long-range dependencies due to reliance on local continuity; (c) Computational Efficiency: Iterative methods (e.g., MLE) incur prohibitive costs for large-scale datasets.

To address these challenges, integration with machine learning technologies is urgently required to overcome dimensionality and complexity constraints.

2.3.2 Machine Learning-Based Methods

The aforementioned traditional methods exhibit several limitations, including poor scalability for large datasets, inability to model complex data distributions, and insufficient accuracy, which motivates the exploration of more effective solutions. Machine learning algorithms offer promising alternatives by discovering latent distributions to address missing data in uncertain scenarios [33].

(1) K-Nearest Neighbors (KNN)

This method estimates missing values by selecting K neighbors based on distance metrics (e.g., Euclidean distance) and computing their weighted average, where weights are inversely proportional to the distance between neighbors and missing data points [34]. Essentially,

observed data from neighboring instances are used to infer missing values [35]. A KNN-based algorithm leveraging spatiotemporal correlations in sensor data has been successfully applied to estimate missing sensor readings [38]. However, this approach suffers from high computational complexity, potentially large estimation errors, and sensitivity to the choice of K value.

(2) Random Forest

Built on ensemble learning principles, random forest employs bootstrap sampling to train multiple independent decision trees on subsampled datasets. By averaging predictions from numerous unpruned classification or regression trees, it effectively forms a multiple imputation scheme. The method's built-in out-of-bag error estimation enables imputation accuracy assessment without requiring separate test data. The missForest algorithm has demonstrated success in handling 10%-30% artificially introduced missing values in mixed-type biological data [36]. Nevertheless, its limitations include: (a) inability to predict values outside the observed range of the imputed variable, which constrains estimation capability; and (b) suboptimal performance for highly skewed data distributions [37].

2.3.3 Deep Learning-Based Methods

While traditional statistical and machine learning approaches demonstrate certain capabilities, they often inadequately capture the temporal dependencies inherent in time-series data [33]. Deep learning methods have shown remarkable performance in handling complex, nonlinear, and large-scale missing data scenarios, making them particularly suitable for sensor data characterized by strong temporal correlations.

In the time-series research community, conventional imputation techniques such as mean value substitution [39] or regression-based methods [40] have been widely adopted. However, these approaches frequently fail to account for nonlinear patterns in the data. The emergence of deep learning techniques for sequential data modeling has introduced advanced neural network architectures including Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks as superior alternatives for time-series imputation [41][42]. These deep learning models excel at preserving nonlinear temporal structures while effectively handling missing values in sensor data streams.

(1) Recurrent Neural Networks (RNNs)

RNNs represent a specialized class of neural networks designed for sequential data processing. By incorporating recurrent structures, RNNs can

capture long-term dependencies in time-series data, making them suitable for predicting consecutive missing values in sensor data streams. The network retains information from previous timesteps and propagates its influence to subsequent outputs, enabling parameter sharing across temporal sequences. However, standard RNNs often suffer from vanishing or

exploding gradients when processing long sequences, which led to the development of advanced variants such as LSTM networks and Gated Recurrent Units (GRUs).

Recent studies have demonstrated the effectiveness of RNNs-based architectures for data imputation. For instance, a bidirectional RNNs with temporal activation regularization was proposed to directly learn and reconstruct missing data patterns [37]. In heterogeneous healthcare platforms under mobile environments—where data collection from multiple devices often introduces redundancy and errors—a multimodal RNNs architecture integrating GRUs with fully connected networks has been developed to address missing value challenges [45].

(2) Long Short-Term Memory (LSTM)

LSTM, an enhanced variant of RNNs, effectively mitigates the vanishing gradient problem inherent in standard RNNs architectures, making it particularly suitable for missing value prediction in long-term temporal sequences. By incorporating specialized gating mechanisms (input, forget, and output gates), LSTM captures both long-term and short-term dependencies within sequential data, enabling robust modeling of temporal patterns in sensor data streams. This capability has proven critical for scenarios requiring accurate reconstruction of missing segments in high-dimensional, time-correlated sensor measurements.

In sensor data prediction tasks, LSTM-based architectures demonstrate superior performance by dynamically retaining and updating contextual information across extended time horizons. For example, studies have leveraged LSTM networks to impute missing values in wearable sensor datasets, achieving significant improvements over traditional interpolation methods through their ability to model complex temporal dynamics.

Recent studies demonstrate LSTM's robustness in handling missing sensor data. For instance, a Deep Convolutional Bidirectional LSTM classifier was benchmarked on sensor datasets with artificially introduced temporal misalignments, achieving 88% and 72% test accuracy in macro- and micro-activity classification, respectively, outperforming state-of-the-art classifiers [44]. While most research focuses on LSTM's tolerance to missing data in classification tasks, some studies explicitly address imputation. A residual LSTM architecture was proposed to predict missing values in the TILES dataset, leveraging hierarchical temporal dependencies to achieve superior reconstruction accuracy compared to conventional methods [43].

(3) Autoencoder

Autoencoder is an unsupervised learning method that can learn the features of data by compressing and reconstructing it [71]. It can be used to fill in the missing values of sensor data and generate data for the missing parts by learning the internal structure of the data.

The training objective of an autoencoder is to minimize the reconstruction error, which is the difference between the original input data and the reconstructed data. Loss functions such as the Mean Squared Error (MSE) are commonly used to measure this difference. During the training

process, a large amount of input data is fed into the autoencoder, and the weights of the neural network layers in the encoder and decoder are adjusted through the backpropagation algorithm to gradually reduce the reconstruction error. As the training progresses, the encoder learns to extract the key features of the data and compress them into an effective code, while the decoder learns to reconstruct the original data as accurately as possible based on these codes[71].

(4) Denoising Autoencoder

DAE, on the other hand, aims not only to reconstruct data but also to recover the original clean data in the presence of noise interference [72]. In most papers, when using DAE to predict missing data, the original data containing missing values or interfering values is inputted, with the goal of restoring or predicting the clean data after filling in the missing values and correcting the interfering values as accurately as possible.

Some people conduct research on childcare activities recognition by using DAE to fill in data and achieve effective accuracy, realizing the suppression of the decline in recognition accuracy when the number of sensors is small[45]. C. Xie et al. propose a Multi-Source Denoising Autoencoder to fill in the missing sensor data in the pedestrian recognition system based on the previously proposed hierarchical AdaBoost. This method constructs a neural network using a Denoising Autoencoder (DAE) to output feature embeddings that include missing sensors, thereby predicting the features of the missing sensor data [47]. N. Jaques et al. introduce a new technique that uses a specialized denoising autoencoder, the Multimodal Autoencoder, to handle missing multimodal data, and demonstrates that it can provide robust emotion prediction even when up to three-quarters of the data sources are lost[48].

2.3.4 Generative Models-Based Methods

A generative model is a type of model that can learn the data distribution and generate new data. By learning from the training data, it understands the latent structure and probability distribution of the data, enabling it to generate

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data similar to the training data. Common generative models include Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), etc.

(1) Variational Autoencoders (VAEs)

The VAE is also composed of an encoder and a decoder. The encoder maps the input data to a latent space, outputting the mean and standard deviation to define a Gaussian distribution (or other probability distributions). Then, through the reparameterization trick, a latent vector is sampled from this distribution, and the decoder converts the latent vector back into data with the same dimension as the original input data. During the training process, the loss function consists of the reconstruction loss and the KL divergence loss. The latter is used to measure the difference between the generated latent probability distribution and the prior probability distribution.

N. U. Okafor et al. studied the performance of Variational Autoencoders (VAEs), Neural Networks with Random Weights (NNRW), Multivariate Imputation by Chained Equations (MICE), Random Forest-based Imputation (missForest), and K-Nearest Neighbours (KNN) in imputing missing values in IoT sensor data. The VAE technique outperformed other methods in calculating missing values with different missing ratios on two real-world datasets [49].

(2) Generative Adversarial Networks (GANs)

According to some studies, GANs outperform classical augmentation methods [53]. A GAN model consists of two main components: the generator and the discriminator. The generator, a deep model, is used to synthesize data, while the discriminator, a binary classifier, is used to detect the generated data from real data. The goal of the generator is to create data similar to real data and deceive the discriminator into recognizing it as real. The training of GANs is based on the min-max theory, where the discriminator and the generator are trained against each other [54]. GANs demonstrate excellent performance in handling the missing data problem of high-dimensional and complex data.

Jinsung Yoon et al. proposed a new GAN-based data imputation method called GAIN [50]. This method estimates the unobserved parts based on the available data, indicating that GAN-based structures can be used to estimate incomplete datasets. Building on previous works [50][51], Li S et al. developed a GAN-based framework named MisGAN to learn the complex high-dimensional distributions of incomplete datasets [52]. The SensoryGAN proposed in the paper successfully created wearable HAR data, and the generated data can be used together with the actual data to improve the accuracy of classifiers [55]. Hu et al. developed a conditional GAN to study the imbalanced data in human activities, attempting to synthesize the three-axis acceleration signals of all activities through a GAN model, with the embedded

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labels of different activities as conditional inputs [56]. The results showed that the synthesized data could solve the problem of imbalanced data and achieved higher accuracy in classifiers. The paper also used a conditional GAN to create synthetic kinematic data, using class labels as additional inputs, and the model trained based on the combined synthetic data achieved more accurate classification results [59]. In another study, Shi et al. proposed a GAN model named HAR-Aug GAN to generate three-axis acceleration, also demonstrating that synthetic data helps improve classification accuracy [57]. To verify the effectiveness of data synthesis for complex activities, the paper proposed using the TheraGAN model to train GAN models separately for the simple activities decomposed from complex activities and for sensors at different body parts [58].

Conditional GANs are relatively commonly used, typically with class labels as additional conditional inputs. However, when faced with data lacking labels or certain application scenarios, there are issues regarding their application.

2.3.5 Multi-Sensor Data Fusion Methods

When data from a certain sensor is missing, data from other sensors can be used for prediction:

(1) Sensor Fusion

Data from multiple sensors or other types of sensors (such as GPS, accelerometers, gyroscopes, magnetometers, etc.) are fused, and the fused data is used to predict missing values.

(2) Multimodal Data

Using deep neural networks to process multimodal data from different sensors can more accurately predict missing sensor data.

In summary, current related studies primarily focus on predicting sensor data with missing values or data segments, typically with a missing ratio controlled within 50% [9]-[49]. However, research on predicting scenarios where wearable sensor data is completely missing remains scarce. Existing studies predominantly use the improvement of activity recognition accuracy as the core evaluation metric. Nevertheless, the application objectives of predicting or synthesizing data are often diverse and not limited to activity recognition alone. Additionally, in application scenarios involving missing data, most machine learning algorithms are dedicated to training classifiers that can adapt to situations with missing data.

In the context of collaborative data collection by multi-location sensors in wearable devices, if the data from a certain sensor location is completely missing, traditional data interpolation methods will fail due to the complete loss of all data information during that time period. In such cases, the essence of the problem shifts to a data generation task, and it is more appropriate to

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use deep learning models or synthetic models to effectively generate the missing data.

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3 Methods and Implementation

3.1 Dataset

The dataset is sourced from a new real-world dataset [60].

3.1.1 Data Introduction

The dataset encompasses various types of sensor data for activities such as climbing downstairs and upstairs, jumping, lying, standing, sitting, running/jogging, and walking, involving fifteen subjects (aged 31.9 ± 12.4 , with a height of 173.1 ± 6.9 cm and a weight of 74.1 ± 13.8 kg, including

eight males and seven females). For each activity, it simultaneously recorded the acceleration data at body positions including the chest, forearm, head, shin, thigh, upper arm, and waist.

Figure 1 Sensors positions [60]

The recording duration for the jumping activity is approximately 1.7 minutes, while the recording duration for the remaining activities is around 10 minutes. The data collection was carried out under real-world conditions, for

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example, subjects walking in the city or jogging in the forest. The collected data includes acceleration data, GPS data, gyroscope data, light data, magnetic field data, and sound level data, etc. Among them, the sampling rate of the recorded acceleration sensors is 50 Hz.

3.1.2 Data Analysis

As previously mentioned, the dataset comprises 7 activity categories involving 15 participants recording by 6 sensors. To facilitate data processing and experimental implementation, this study exclusively extracted data from accelerometer sensors (encompassing three feature dimensions: X, Y, and Z) for analysis. Among the eight activities, completely static activities and the somewhat unstable jumping action were excluded, and only four activities with periodic patterns, namely walking, running, climbing up, and climbing down, were retained. In the raw data, each activity recording has a duration of approximately 10 minutes (500–600 seconds), with the longest recordings spanning 12.66 minutes, corresponding to a data length of approximately 30,000 points (calculated as 600 seconds $\times$ 50 Hz). However, data preprocessing revealed significant missing data issues: in sub2's "climbing up" activity, the data from one complete accelerometer sensor were entirely missing; meanwhile, during "running" activities, the forearm sensor data for multiple participants exhibited approximately half-duration missing values. Given the substantial amount of missing data, the final experimental dataset excluded sub2's "climbing up" recordings and the running activity samples with forearm sensor deficiencies, retaining only the valid data from the remaining 14 participants. As a result, the shape of the sample should look like 30000 (the data length

of each person's each activity recording by each position sensor) $\times$ 3 (feature dimensions: X, Y, and Z) and the number is 14 (number of people) $\times$ 4 (number of activity) $\times$ 6 (number of sensors).

3.1.3 Data Preprocessing

In line with the suggestions in the article, data was segmented using windows with a length of approximately 1 second (64 data points) and a 50% overlap [59]. Therefore, each 10-minute long sequence of data can be segmented into more than 1000 samples with a length of 64. And the

sample of different position sensor from the same person and activity would be combined together. As a result, after segmentation and combination, the shape of each sample becomes $64(\text{length of the data}) \times 21(3 \times 7)$, in which, 7 is the number of sensors) and the number of samples is $15 \times 4 \times 1000$.

For the experiment, the data from the sensor at the chest was selected as the target missing data for prediction. There are totally around 60000 timeframes. For each person and activity, the former 70% samples would be divided as training set while the last 30% samples would be divided as testing set. So the ratio of training set and testing set is 7:3.

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3.2 Proposed Architecture

This section presents the detailed information of all the method architectures used in this paper. Regarding the model's input data, as previously mentioned, each sample originally contains 21 features (3 feature dimensions: X, Y, and Z $\times$ 7 sensors). During training, the chest sensor data is excluded, resulting in a reduced feature dimension of 18 (3×6) for the training dataset.

3.2.1 Autoencoder

As mentioned in the related work, AE is composed of a decoder and an encoder. The AE used in the experiment is mainly constructed by convolutional layers. It takes the successfully collected sensor data as input. Through training, it reduces the difference between the output sensor data and the original data, and recovers the complete sensor data (that is, including the missing sensor data).

Figure 2 Structure of AE

Figure 2 illustrates the architecture of an Autoencoder. The process of this architecture starts with the original data. The complete feature data of the sensors from the corresponding body parts are selected as the training data for learning. The training data is fed into the Encoder part of the autoencoder. The encoder consists of an input layer and an intermediate hidden layer. After being processed by the encoder, the data is transformed into a low-dimensional representation, which then enters the Decoder. The decoder is composed of an intermediate hidden layer and an output layer, and it outputs the generated data that includes the features of all sensors from various parts. The entire training process of the autoencoder uses the Mean Square Error as the loss function. By minimizing the mean square error between the generated data and the original data, the weights of the encoder and decoder are adjusted, thus achieving effective data learning.

The detailed structure of a Conv-AE (Convolutional Autoencoder) based on convolutional and deconvolutional layers for the missing data imputation problem is shown in Figure 3.

Figure 3 Structure Detail of AE

The input data exhibits a shape characteristic of (64, 18, 1), where 64 represents the sequence length and 18 represents the data dimension. These data are obtained from the acceleration sensors of 6 known body parts. Each sensor has three-axis features, thus constituting 18-dimensional data. In the encoder section, four sets of combined structures consisting of convolutional layers and max-pooling layers are designed. The size of the convolutional kernels used in each convolutional layer is uniformly set to (8, 3), and the number of convolutional kernels starts from 64 and gradually decreases to 8 in a halving pattern. Specifically, after each convolutional operation, the ReLU activation function is applied to introduce non-linearity and enhance the model's representational ability. Meanwhile, due to the unique nature of the data, the size of the pooling window for each max-pooling operation is

adjusted accordingly to ensure that key information is not overly lost while effectively extracting features. In addition, to effectively prevent the potential problem of gradient explosion during model training, a Batch Normalization layer is added between each convolutional layer and max-pooling layer. This layer normalizes the input of each layer, making the model training more stable and efficient. The decoder section has a structure corresponding to that of the encoder.

The convolutional layers in the encoder are replaced with transposed convolutional layers, which can perform upsampling operations, forming an inverse operation to the downsampling operations (implemented by max-pooling layers) in the encoder. Meanwhile, the max-pooling layers in the encoder are converted into upsampling layers, and the size of the sampling window is flexibly and reasonably adjusted according to the characteristics of the input data and the target output data to ensure that the target data dimension can be accurately obtained. Finally, a convolutional layer with 1 convolutional kernel is set to compress and aggregate the data, and Tanh is selected as the activation function for this layer. The final output data has a shape of (64, 21, 1).

3.2.2 Variational Autoencoder

As mentioned in the related work, there are significant differences in the output mechanisms of the encoders between the Autoencoder (AE) and the Variational Autoencoder (VAE). The encoder of the AE maps the input data to a definite low-dimensional code, while the encoder of the VAE outputs a set of statistical parameters, namely the mean and the standard deviation, which are used to define a probability distribution in the latent space, typically a Gaussian distribution. This output method of probability distribution enables the VAE to capture the latent uncertainties in the data,

rather than merely obtaining a single feature representation. After obtaining the parameters of the probability distribution, the VAE samples the latent vector from this distribution through the reparameterization trick. Specifically, the reparameterization trick transforms the sampling process from non-differentiable random sampling into differentiable mathematical operations, allowing the entire network to be trained end-to-end using the backpropagation algorithm. For example, given the mean (μ) and the standard deviation (σ), the latent vector (z) can be calculated as

$z = \mu + \sigma \cdot \varepsilon$, where ε is a random vector sampled from the standard normal distribution, and $*$ represents element-wise multiplication. In the decoder part, the VAE uses the latent vector sampled from the probability distribution for data reconstruction. The decoder maps the latent vector back to a space with the same dimension as the original input data, attempting to restore the features and structure of the original data. During the training process, the loss function of the VAE consists of two parts: one is the

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reconstruction loss, which usually uses the Mean Squared Error (MSE) or cross-entropy loss to measure the difference between the reconstructed data and the original data, ensuring that the model can accurately restore the input information; the other is the KL divergence loss [73], which is used to measure the difference between the latent probability distribution output by the encoder and the prior probability distribution (usually the standard normal distribution). By minimizing the KL divergence loss, the VAE can constrain the distribution in the latent space, avoiding the model from learning overly complex or meaningless distributions, thus improving the model's generalization ability and the diversity of generated data.

Figure 4 Structure of VAE

The detailed structure is shown in Figure 5.

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Figure 5 Structure Detail of VAE

The structure and configuration are almost identical to those of the AE. The key difference lies in the need to sample the latent vector from the probability distribution of the latent vector through the reparameterization trick.

3.2.3 CNN-LSTM

As discussed in the related work, Convolutional Neural Networks (CNNs), by virtue of their local connectivity and weight-sharing properties, demonstrate remarkable superiority in feature extraction tasks for data such as images and signals. They can efficiently capture the spatial structural information of data. Long Short-Term Memory networks (LSTMs), as an improved version of Recurrent Neural Networks (RNNs), effectively address the problems of gradient vanishing and gradient explosion that occur in traditional RNNs through their unique gating mechanisms. LSTMs are particularly adept at handling sequential data with temporal dependencies and are capable of modeling the dynamic changes in data over time.

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In this study, the training data is sourced from sensors at 6 other body parts, covering information with 6×3 dimensions, while the target missing data is from the sensors at the chest, with 1×3 dimensions. Since the data to be pre-dicted has sequential characteristics, a single model is insufficient to meet the dual requirements of spatial feature extraction and temporal feature learning simultaneously. Therefore, we selected an architecture that combines CNN and LSTM for missing data prediction. Specifically, as the front-end network, CNN can compress the multi-dimensional data collected by the 6 sensors. Through the alternating operations of convolutional layers and pooling layers, it extracts the spatial features of the data and explores the potential correlations among different sensor dimensions. The LSTM, as the back-end network, receives the feature vectors output by the CNN and processes them as sequential data. Through its internal forget gates, input gates, and output gates, LSTM can selectively retain or discard historical information, thereby learning the patterns and dependencies of sensor data in the temporal dimension. Through the collaborative operation of CNN and LSTM, the effective extraction and utilization of spatial and temporal features in complex sensor data are achieved, laying a foundation for the accurate prediction of missing data.

Figure 6 Structure of CNN-LSTM

As shown in Figure 6, the network structure of CNN-LSTM includes a convolutional network part and a sequential data processing part. The convolutional network part is composed of two combinations of convolutional layers and max-pooling layers. The size of the convolutional kernels is (3, 3), and the number of convolutional kernels doubles successively. The first max-pooling layer first compresses and extracts the features of each sensor, and the second pooling layer then uniformly compresses and extracts the features of all sensors. After reshaping into 64 time series, the data enters the LSTM part for time-series feature learning and extraction. Finally, the data dimensions are reshaped into (64, 3, 1) through the Time distributed layer and the Reshape layer.

Figure 7 Structure Detail of CNN-LSTM

3.2.4 cGAN

Conditional Generative Adversarial Networks (cGAN), as a crucial improvement paradigm of standard Generative Adversarial Networks (GANs), demonstrate unique application potential in the field of multi-sensor data processing for wearable devices. Compared with the unconstrained data generation mode of traditional GANs based on random noise, cGAN introduces external conditional information to construct a condition-driven data generation mechanism. In this study, multi-dimensional data collected from other sensors are used as conditional inputs to achieve precise generation of missing target sensor data. From the perspective of network architecture, the generator of cGAN fuses the random noise vector with the effective data feature vector of other sensors as the input of the generation model. Specifically, when the chest sensor data is missing, multi-dimensional data, such as acceleration and gyroscope data, collected from six body parts including the arms and legs (a total of 6×3 -dimensional features) serve as conditional information. Together with the random noise, they guide the generator to learn the mapping relationships

among different sensor data. Through operations such as convolution and deconvolution, the generator outputs missing data that match the dimensions of the target sensor (chest sensor, 1×3 -dimensional features). The discriminator, on the other hand, receives real/generated data and conditional data simultaneously, constructing a binary classification model based on conditional constraints. By judging the rationality of the data under given conditions, it realizes the discrimination of the authenticity of the generated data. During the training process, cGAN follows the adversarial learning mechanism, and the generator and discriminator achieve collaborative optimization through a min-max game. The generator aims to utilize the conditional data from other sensors to generate target sensor data consistent with the distribution of real data, maximizing the misjudgment probability of the discriminator; the discriminator enhances its ability to distinguish generated data by combining conditional information. The loss function is composed of data reconstruction loss and conditional adversarial loss. The reconstruction loss measures the feature difference between the generated data and the real data, while the conditional adversarial loss constrains the data generation ability of the generator under specific conditions. By minimizing the joint loss function through the backpropagation algorithm, the generator is prompted to deeply explore the potential correlations among different sensor data, effectively completing the missing data. In the scenario of multi-sensor collaborative collection of wearable devices, the cGAN architecture with data from other sensors as conditions has significant advantages. When the data of a single sensor is completely or partially missing, this method can utilize the redundant information of other sensors to generate target data that conforms to spatio-temporal

characteristics through conditional constraints, effectively solving the failure problem of traditional interpolation methods in scenarios of complete data loss. Meanwhile, this mechanism can be further applied to the enhancement of imbalanced datasets. By generating missing data samples under specific conditions, it optimizes the data distribution, providing a high-quality data foundation for subsequent tasks such as activity recognition and behavior analysis based on complete datasets, and improving the generalization performance and prediction accuracy of models.

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Figure 8 Structure of cGAN

As shown in Figure 8, the generator directly employs the CNN-LSTM network structure. However, the input data is different. The 18-dimensional training data is used as an additional conditional input to the generator, which is concatenated with 3-dimensional noise to predict the three-axis data collected by the chest acceleration sensor. The loss function used by the generator is also known as the reconstruction loss, which is the MSE value between the generated data and the real data. The generator needs to continuously train to minimize the reconstruction loss. To prevent the discriminator from becoming too powerful, a simple convolutional neural network is selected to determine whether the data is generated or real. The loss for the discriminator's learning is the adversarial loss, which is a way to measure the similarity between the data generated by the generator and the real data. The set adversarial function consists of two terms: one measures the probability of predicting real data as real, and the other measures the probability of predicting generated data as fake. The total loss of the discriminator is the sum of the loss for real data and the loss for generated data. Therefore, the higher the loss, the better the discrimination effect of the discriminator. Similarly, this loss also needs to be minimized.

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Figure 9 Structure Detail of cGAN

As shown in Figure 9, compared with the previous CNN-LSTM network structure, the input data of the generator has changed. The 18-dimensional features are concatenated with 3-dimensional noise. Therefore, the pooling window of the pooling layer has been adjusted accordingly. The discriminator is mainly composed of two convolutional layers and a max-pooling layer. Finally, a probability value is output by using the Sigmoid activation function.

3.3 Training Process

As mentioned in Data Preprocessing part, 70% original samples are divided into training set while 30% of them are testing set. The number of features in each sample is 3×7 . As the data collected from chest sensor is regarded as goal data to predict, the input shape of training data is $64 \times (3 \times 6)$.

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3.3.1 Early Stop

To prevent the model from overfitting and save training time and computational resources, an early stopping mechanism is implemented during the training process. The core idea of this mechanism is to monitor a certain loss metric (such as the Mean Squared Error, MSE) during training. When this metric shows no significant improvement over a certain number of epochs, it is determined that the model has reached a satisfactory training state, and subsequent training is terminated. Specifically, no significant improvement means that the current loss value is not smaller than the best loss value minus a user-defined minimum change amount. The number of epochs that fail to meet the expected improvement is recorded.

3.3.2 Optimizer

Adam is selected as the optimizer for all experiments with default parameter settings. Adam is chosen because it can adaptively adjust the learning rate, accelerate convergence by incorporating momentum, avoid local minima, and is applicable to various types of data and models. Additionally, it features fast convergence speed and good stability.

3.3.3 Generator and Discriminator

After multiple trials, due to the relatively fast convergence speed and strong learning ability of the discriminator, its parameters are fixed for the first 100 epochs out of a total of 500 epochs. Subsequently, the two networks are trained jointly, but the training rule is set such that the generator is trained twice for each training of the discriminator until convergence is achieved.

3.4 Evaluation Metrics

3.4.1 Root Mean Squared Error (RMSE)

RMSE quantifies the differences between predicted and true values, and is widely adopted in regression analysis. Mathematically, it is expressed as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

n

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

n

$i=1$

where n represents the number of samples, y_i is the true value of the i -th sample, and $\hat{y}_i$ is the corresponding predicted value. A lower RMSE indicates higher prediction accuracy. By squaring the differences, RMSE emphasizes larger errors, making it sensitive to outliers in the data.

3.4.2 Mean Absolute Error (MAE)

MAE measures the average magnitude of the errors between predictions and true values without considering their direction. Its formula is given by:

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MAE

n

$\sum |y_i - \hat{y}_i|$

n

$i=1$

Unlike RMSE, MAE treats all errors equally since it does not square the differences. This characteristic makes MAE less influenced by extreme values and provides a more straightforward representation of the average prediction error. In scenarios where the impact of outliers needs to be minimized, MAE serves as a more robust evaluation metric.

3.4.3 Mean Squared Error (MSE)

MSE calculates the average of the squared differences between predicted and true values, defined as:

MSE

n

$\sum (y_i - \hat{y}_i)^2$

n

$i=1$

As a fundamental loss function in regression model training, MSE penalizes larger errors more severely due to the squaring operation. This property drives the model to prioritize reducing significant discrepancies during the training process, encouraging it to find optimal parameter combinations that better fit the data distribution. By minimizing MSE, the model can achieve higher prediction accuracy and improved generalization ability, making it a crucial metric for model optimization.

3.4.4 Coefficient of Determination

R^2 evaluates how well the regression model fits the observed data, with its value ranging from 0 to 1:

$R^2 =$

$$\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$i=1$

$$\sum_{i=1}^n (y_i - \bar{y})^2$$

$i=1$

where $\bar{y}$ is the mean of the true values. A value of R^2 close to 1 indicates that the model explains a large proportion of the variance in the data, suggesting an excellent fit. Conversely, a value close to 0 implies that the model performs poorly and fails to capture the underlying patterns in the data effectively.

3.4.5 Correlation Coefficient

The correlation coefficient measures the linear dependence between pre-dicted and true values. For a sample of size n , the Pearson correlation coefficient is calculated as:

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CC

$$\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

$$i=1$$

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2$$

$$i=1$$

where $\bar{x}$ and $\bar{y}$ are the means of the true and predicted values, respectively. The coefficient ranges from -1 to 1. A value close to 1 or -1 signifies a strong positive or negative linear relationship, respectively, indicating accurate predictions. A value near 0 implies weak or no linear correlation, suggesting that the model's predictions may not closely match the actual values.

3.4.6 Signal-to-Noise Ratio (SNR)

In the context of evaluating prediction models, especially for time-series or signal-related data, SNR is defined as:

$$SNR = 10 \log_{10} \left(\frac{\sum_{i=1}^n y_i^2}{\sum_{i=1}^n (y_i - \hat{y}_i)^2} \right)$$

$$\sum_{i=1}^n y_i^2$$

$$i=1$$

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$i=1$$

A higher SNR indicates that the predicted values closely resemble the true signal, meaning the model effectively captures the underlying patterns. Conversely, a lower SNR suggests that the model's predictions contain substantial errors or are affected by noise, highlighting the need for model improvement or further data preprocessing.

4 Results

4.1 Effect of Different Models

In this experimental study, four model architectures were employed: the Autoencoder (AE), Variational Autoencoder (VAE), CNN-LSTM hybrid model (integrating Convolutional Neural Networks and Long Short-Term Memory Networks), and Conditional Generative Adversarial Network (cGAN). While both AE and VAE generated complete sensor data across seven anatomical sites, model evaluation focused exclusively on predicting data from the target site. Specifically, only the predicted values corresponding to the target-site sensors were utilized for calculating all evaluation metrics. This targeted approach ensures rigorous quantification of model performance with respect to the specific task, emphasizing relevance to the region of interest.

Table 3 Results based on different model architectures

	RMSE	MSE	MAE	R2	CC	SNR
--	------	-----	-----	----	----	-----

AE	3.272	10.707	2.224	0.709	0.984	3.919
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VAE	3.300	10.895	2.299	0.704	0.988	3.843
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CNN-LSTM	2.783	7.743	1.756	0.780	0.995	14.433
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cGAN	2.986	8.916	1.909	0.751	0.988	10.621
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Based on the data in the table, the Autoencoder (AE) model outperformed the Variational Autoencoder (VAE) model in terms of error control, data correlation, and signal-to-noise ratio (SNR). This superiority may stem from AE's direct learning of dominant data features during the encoding-decoding process, enabling more effective prediction of target-site sensor data. In contrast, VAE's probabilistic generative model introduces inherent uncertainty during data generation, which likely contributes to its slightly lower prediction accuracy compared to AE. The CNN-LSTM hybrid model demonstrated the best overall performance. This can be attributed to the complementary strengths of its components: Convolutional Neural Networks (CNNs) excel at extracting spatial features from sensor data, while Long Short-Term Memory (LSTM) networks efficiently capture temporal dynamics in sequential data. Their integration allows the model to learn the spatiotemporal motion patterns and inter-site correlations of human body movements, thereby achieving high precision in predicting missing sensor data. The Conditional Generative Adversarial Network (cGAN) built upon the CNN-LSTM framework showed marginally inferior performance, with slightly higher errors and lower correlation

coefficients than CNN-LSTM. Notably, both models significantly outperformed the AE family. This suggests that while the adversarial training mechanism in cGAN enhances model expressiveness

and generalization, it also increases architectural complexity and training difficulty, potentially limiting its ability to fully leverage the underlying CNN-LSTM features in certain scenarios. The CNN-LSTM model's highest SNR indicates its superior capability to focus on informative data features during learning and prediction, minimizing the influence of noise. In contrast, the features learned by AE and VAE appear less reliable, likely due to their inability to capture the spatiotemporal correlations in complex human motion data as effectively as CNN-LSTM, making them more susceptible to noise interference during prediction. Additionally, both VAE and GAN-based models underperformed their deterministic counterparts (e.g., AE and CNN-LSTM) that rely purely on original data features. This trade-off may arise from their need to balance data generation quality with distributional consistency, which inherently compromises prediction precision to some extent.

4.2 Effect of Different Actions

Table 4 Results of AE based on Different Actions

	RMSE	MSE	MAE	R2	CC	SNR
Climbing Down	2.785	8.596	2.078	0.649	1.315	5.020
Climbing Up	2.274	5.699	1.788	0.748	1.456	7.245
Running	4.177	20.518	3.143	0.595	0.957	7.932
Walking	2.082	4.660	1.604	0.829	1.294	8.416

Table 5 Results of VAE based on Different Actions

	RMSE	MSE	MAE	R2	CC	SNR
Climbing Down	2.851	8.938	2.179	0.636	1.30	4.410
Climbing Up	2.259	5.455	1.787	0.754	1.462	7.171
Running	4.230	20.805	3.230	0.587	0.965	5.743
Walking	2.166	4.994	1.710	0.816	1.289	7.035

Table 6 Results of CNN-LSTM based on Different Actions

	RMSE	MSE	MAE	R2	CC	SNR
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Climbing Down	2.485	7.051	1.833	0.709	1.318	6.635
Climbing Up	1.959	4.386	1.528	0.800	1.420	10.680
Running	3.376	15.157	2.440	0.687	1.006	12.216
Walking	1.490	2.372	1.117	0.911	1.302	15.455

Table 7 Results of cGAN based on Different Actions

	RMSE	MSE	MAE	R2	CC	SNR
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Climbing Down	2.611	7.660	1.909	0.682	1.340	5.231
Climbing Up	2.128	5.240	1.645	0.768	1.383	7.862
Running	3.683	17.211	2.651	0.658	0.997	8.056
Walking	1.708	3.133	1.278	0.884	1.296	10.938

As shown in the Table 4-7, the "running" action exhibited the poorest error control and the lowest correlation with original data across all models. Conversely, the "walking" action demonstrated the best error control and highest data correlation. This discrepancy may be attributed to the inherent complexity of running, which involves rapid coordination of multiple joints and drastic acceleration changes. Such dynamic characteristics make it challenging for models to accurately capture the motion patterns during training and prediction. In contrast, walking—a highly periodic and regular motion—features relatively simple and stable movement patterns, enabling models to more effectively learn its characteristic features and achieve superior prediction performance. Additionally, the walking action recorded the highest signal-to-noise ratio (SNR), indicating that its associated sensor data contains minimal interfering noise. This allows models to extract informative features more efficiently while minimizing noise-induced errors. Conversely, the "climbing down" action had the lowest SNR values, likely due to significant changes in body posture and complex relative motions between sensors during this activity. External factors such as handrail interactions and stair geometry may further introduce extraneous noise into the data. As a result, models processing climbing down data are more prone to noise interference, potentially reducing the reliability of their prediction outcomes.

4.3 Partial Visualization of Predictions

In this section, we further demonstrate the models' prediction performance through visualizations.

Figure 10 Partial Visualization Result of AE

Figure 11 Partial Visualization Result of VAE

Figure 12 Partial Visualization Result of CNN-LSTM

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Figure 13 Partial Visualization Result of cGAN

Analysis of the generated frame sequence (sub12, walking activity) reveals distinct performance differences among the models. Both the CNN-LSTM and cGAN models successfully capture the inherent fluctuation trends within the gait data. In contrast, the DAE and VAE models fail to adequately model these dynamics, likely due to their simpler architectures and lack of dedicated sequential processing capabilities. Their generated sequences often appeared oversimplified and lacked the necessary temporal coherence. Key limitations likely contributing to this inferior performance include their fundamentally weaker sequential modeling capabilities. DAE primarily focus on feature extraction and reconstruction within static or pointwise contexts, lacking dedicated mechanisms to learn long-range dependencies in time series data. Similarly, while VAE learns a probabilistic latent representation, their standard formulations often generate independent frames or very short sequences, failing to effectively model the complex, high-order temporal dependencies inherent in continuous gait motion. Furthermore, the data generated by the CNN-LSTM model exhibits greater smoothness compared to the output of the cGAN. The cGAN output, although capturing the overall dynamics, occasionally displayed minor temporal discontinuities or "jitter", potentially stemming from the nature of its adversarial training or the discrete sampling inherent in frame generation without explicit strong temporal constraints. In conclusion, CNN-LSTM and cGAN models with explicit temporal modeling capabilities can effectively capture gait dynamic features, while DAE and VAE models lacking dedicated sequence processing mechanisms fail in modeling complex temporal dependencies and generate dynamically incoherent simplified sequences. Although both CNN-LSTM and cGAN can reconstruct the gait fluctuation trend, CNN-LSTM generates significantly smoother and more continuous results with its inherent sequence optimization mechanism (LSTM), while the output of cGAN is limited by the inherent attributes of adversarial training and is prone to slight temporal discontinuity.

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4.4 Effect of Different Position Data Selected

Given that the current prediction framework relies on sensor data from allother 6 anatomical sites to infer missing target-site data, a critical questionarises: is it necessary to incorporate data from so many sites during training?This issue warrants further investigation, particularly in application scenar-ios such as virtual live streaming, where full-body motion capture deviceswith multiple sensors are used. If sensor data from certain body parts can beaccurately predicted using data from nearby sites, the motion capture systemcould be significantly lightweighted by reducing the number of physical sen-sors.To solve this problem, a follow-up experiment was designed using only datafrom two neighbouring sensors at the target location as training input to pre-dict missing chest sensor data. Head and waist sensors were selected as theoptimal neighbouring sources due to two key factors:(1) Anatomical proximity [69]: both sensors reside in the core torso region(like the chest), exhibiting stronger motion correlation with the chest thandistal limbs;(2) Kinematic coupling [70]: chest movements (anterior-posterior/lateral)inherently drive synchronized adjustments in head posture and waist cen-troid shifts, forming a robust dynamic relationship.This selection ensures feature relevance while avoiding high-noise interfer-ence from extremity sensors. Comparative tests were conducted using iden-tical model architectures to validate this design choice.

Table 8 Results based on Different Position Data Selected

	RMSE	MSE	MAE	R2	CC	SNR
AE	5.111	26.122	3.482	0.291	0.735	1.608
VAE	4.914	24.143	3.290	0.345	0.806	1.740
CNN+LSTM	5.122	26.237	3.465	0.156	0.693	8.991
cGAN	5.321	28.315	3.527	0.125	0.675	7.210

As indicated in Table 8, all models exhibited significantly degraded perfor-mance compared to their full-site counterparts when trained solely on headand waist sensor data. Notably, the CNN-LSTM model—previously the topperformer—and the cGAN model showed marked declines in loss control andcorrelation coefficients, even falling below the AE and VAE models. This deg-gradation stems from the limited kinematic information available from onlytwo neighbouring sensors (head/waist), which restricts the models' ability tocapture the complex inter-site dependencies essential for accurate chest

motion reconstruction. The substantial accuracy drop under this reduced-data scenario underscores the critical role of multi-sensor fusion in preserv-ing motion dynamics.Surprisingly, the VAE model exhibited the best prediction accuracy and cor-relation in this setup, despite its

probabilistic generative framework being originally designed for uncertainty tolerance rather than deterministic reconstruction. This counterintuitive result suggests that VAE's latent space modelling effectively leverages the statistical distribution of torso-centric motion patterns (head/waist/chest). In contrast, CNN-LSTM and cGAN—despite ranking second-to-last in accuracy—maintained top-two positions in signal-to-noise ratio (SNR), indicating their discriminative feature extraction capabilities prioritize noise suppression over structural fidelity when facing data scarcity. These findings reveal a fundamental trade-off between feature completeness and model robustness in sensor-reduced scenarios. While multi-sensor fusion remains ideal for maximizing prediction accuracy, the VAE's performance highlights potential for lightweight systems prioritizing probabilistic plausibility over deterministic precision. Future work should explore hybrid architectures that dynamically balance feature dependency modelling with noise-resistant representations under variable sensor configurations.

4.5 Effect of Prediction based on Different Individuals

To evaluate the cross-individual generalization ability of the model, a strict dataset segmentation scheme based on identity-exclusive was designed: 4 individuals were randomly selected from the data of all 14 individuals as the unknown test set, and the data of the remaining 10 individuals were all included in the training set. This division ensures that the test individuals are completely invisible during the training phase, thereby effectively testing the model's adaptability to the heterogeneity of unknown individuals. Based on this partitioned dataset scenario, the results of each model are shown in Table 9.

Table 9 Results based on Different Individuals

	RMSE	MSE	MAE	R2	CC	SNR
AE	6.282	39.467	4.686	-0.164	0.511	1.063
VAE	6.647	44.176	5.032	-0.302	0.409	0.949
CNN-LSTM	6.422	41.242	4.681	-0.314	0.564	3.059
cGAN	6.153	37.861	4.426	-0.179	0.617	3.354

Although cGAN outperforms other models in terms of CC (0.617) and SNR(3.354), its RMSE (37.861) and MAE (4.426) still demonstrate insufficient generation quality. This result confirms that with a limited data scale of only 14 individuals, none of the existing architectures can overcome the heterogeneity among individuals (such as differences in bone length and stride frequency). The essence is that the training set fails to cover a sufficient variety of identity features, resulting in the model being able to only remember specific individual patterns and losing its generalization ability.

5 Summary

5.1 Discussion

5.1.1 Superior Performance of CNN-LSTM and cGAN

The experimental results conclusively demonstrate CNN-LSTM as the optimal model for missing sensor data prediction, achieving the highest accuracy across all metrics. This superiority stems directly from its dual-branch spatiotemporal architecture: (1) Convolutional layers learn spatial correlations across anatomical sites, extracting invariant features from multi-sensor inputs; (2) LSTM modules capture temporal dynamics, modelling motion continuity via memory gates. This synergistic design enables robust reconstructions even during complex actions like running. While cGAN ranks second, its adversarial training mechanism provides enhanced generative diversity, though occasionally inducing minor temporal artifacts ("jitter" in Fig. 13) due to optimization instability. Crucially, both models significantly outperform AE/VAE, whose frame-wise processing paradigms collapse under motion dynamics, failing to disentangle noise from physiological signals ($\text{SNR} < 4$).

5.1.2 Model Structure-Behaviour Relationships

The consistent performance hierarchy observed within architectural families—AE surpassing VAE and CNN-LSTM outperforming cGAN—reveals intrinsic trade-offs between deterministic reconstruction and probabilistic generation. (1) AE vs. VAE: Precision versus Regularization While both employ encoder-decoder frameworks, AE's direct feature learning minimizes reconstruction error through deterministic mapping. In contrast, VAE's probabilistic latent space enforces distributional constraints (e.g., KL divergence), injecting irreducible uncertainty during sampling. This fundamentally compromises its prediction fidelity, manifesting as blurred temporal details and marginally lower R^2 values across all actions. (2) CNN-LSTM vs. cGAN: Stability versus Expressiveness Both architectures share spatiotemporal modelling capabilities, yet CNN-LSTM's sequential optimization (via backpropagation through time) ensures frame-to-frame coherence. Meanwhile, cGAN's adversarial training—though enhancing motion diversity—amplifies high-frequency artifacts through discriminator-guided noise injection. This explains its slightly elevated error rates and occasional temporal jitter.

5.1.3 Impacts of Reduced Sensor Data

When trained with only two neighbouring sensor inputs, all models experienced significant performance degradation. Notably, VAE achieved the highest prediction accuracy under this constraint, likely due to its probabilistic

generative nature, which enables learning data distributions to produce plausible predictions even with limited inputs. This suggests that VAE may offer unique advantages in scenarios with restricted data availability, such as sensor-constrained lightweight systems. In conclusion, this study solves the problem of data discontinuity caused by the absence of sensors in the practical application of human motion tracking technology by proposing a sensor data generation framework based on deep learning. By systematically comparing the performance of autoencoders (AE), variational autoencoders (VAE), CNN-LSTM hybrid models, and conditional generative adversarial networks (cGAN) under different modes (full sensor distribution/local sensor distribution/multi-action/cross-individual), the core findings are as follows: (1) The decisive role of spatio-temporal modeling The CNN-LSTM model, relying on the convolution - cyclic dual-path architecture (spatial feature extraction + temporal dynamic modeling), achieves the optimal prediction performance under complete sensor conditions, confirming the precise capture of the inherent laws of human movement. (2) Paradigm shift in resource-constrained scenarios When trained only with data from adjacent sensors, VAE demonstrates unique advantages. Its potential spatial distribution learning mechanism can generate reasonable outputs from incomplete inputs, providing a technical reference path for lightweight systems and enabling more in-depth experiments based on this. (3) Systematic failure of cross-individual generalization All models completely collapsed in tests without seeing the subjects. This failure was due to the difficulty of the training set in covering individual heterogeneity, exposing the flaws of the algorithm.

5.2 Future Works

5.2.1 Model Optimization and Enhancement

Further optimize the network architectures and training algorithms of CNN-LSTM and cGAN to improve their prediction accuracy and robustness in handling complex motion data. For instance, incorporating attention mechanisms could enable the models to focus on critical features within the data, thereby enhancing prediction performance. This approach may mitigate the information loss caused by spatiotemporal feature aggregation in hybrid networks.

5.2.2 Real-Time Performance Research

Conduct research on the real-time capabilities of the models to meet the latency requirements of practical applications. In scenarios such as virtual reality and real-time motion analysis, where real-time performance is critical, model compression techniques (e.g., pruning, quantization) and algorithmic

optimizations (e.g., parallel computing frameworks) can be employed to enhance inference efficiency. The goal is to enable instantaneous sensor data prediction and supplementation in real-world deployments.

5.2.3 Non-Periodic and Sudden Motion Analysis

Expand the research scope to non-periodic and sudden movements, exploring prediction models and methodologies tailored to these motion types. Due to their complex and variable patterns, such motions pose higher challenges to model generalization. Adopting adaptive deep learning architectures (e.g., transformers with dynamic attention) or integrating advanced machine learning techniques (e.g., meta-learning for rapid adaptation) may be necessary to address the unpredictability of such motions.

5.2.4 Sensor Deployment Optimization

Building on the insights into the impact of sensor density and spatial topology revealed in this study, further investigate optimal sensor deployment strategies. Through systematic layout design, the aim is to minimize the number of sensors and energy consumption while maintaining motion capture accuracy, thereby achieving lightweight optimization of motion capture systems. This may involve graph-based topology modelling to identify critical sensor nodes and reduce redundant deployments.

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Appendix A

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Core Codes:<https://github.com/hhhen/SensorDataPredict.git>

<https://www.uni-mannheim.de/dws/research/projects/activity-recognition/dataset/dataset-realworld/>

<https://github.com/hhhen/SensorDataPredict.git>